

Dataset

1

- ASVspoof 2019 LA
- Bona fide / Spoof
- Known and unseen attacks



Preprocessing

2

- Mono conversion
- 16 kHz resampling
- Waveform normalization
- Padding and masking



Augmentation

3

- RawBoost ① + ②
- Codec augmentation
- Real-world distortions



Teacher Model

4

- Wav2Vec2-XLSR front-end
- AASIST back-end
- P2SGrad / OC-Softmax loss



Soft Predictions

5

- Teacher confidence scores
- Soft labels for training



Student Model

6

- Frozen Wav2Vec2-XLSR
- AASIST-L lightweight back-end



Knowledge Distillation

7

- Hard label loss
- KD loss
- Teacher knowledge transfer



Prediction

8

- Bona fide
- Spoof



Evaluation

9

- EER and min-tDCF
- Accuracy, precision, recall
- F1-score and confusion matrix